

CHAPTER 6

Kramers–Kronig Relations

General relation between the real and imaginary parts of $\tilde{n}$ or ε

- Now just go to time and real space .
- The most general expression for a **linear response** function is :

$$\frac{1}{\varepsilon_0} \mathbf{P}(\mathbf{r}, t) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \chi(\mathbf{r}, \mathbf{r}', t, t') \mathbf{E}(\mathbf{r}', t') dt' d\mathbf{r}' \quad (6.1)$$

The polarization $\mathbf{P}$ at point $\mathbf{r}$ and time t depends on the electric field at all other places and at all times

First, we assume the sample is homogeneous in time, χ depends only on $t - t'$;

Second, $\mathbf{E}(\mathbf{r}', t')$ are much longer than the lattice constant, depends only on $\mathbf{r} - \mathbf{r}'$:

$$\frac{1}{\varepsilon_0} \mathbf{P}(\mathbf{r}, t) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \chi(\mathbf{r} - \mathbf{r}', t - t') \mathbf{E}(\mathbf{r}', t') dt' d\mathbf{r}' \quad (6.2)$$

$$\varepsilon_1(\omega) - 1 = \text{Re} \{ \chi(\omega) \} = \frac{2}{\pi} P \int_0^{+\infty} \frac{\omega' \varepsilon_2(\omega')}{\omega'^2 - \omega^2} d\omega'$$

$$\varepsilon_2(\omega) = \text{Im} \{ \chi(\omega) \} = -\frac{2\omega}{\pi} P \int_{-\infty}^{+\infty} \frac{\varepsilon_1(\omega') - 1}{\omega'^2 - \omega^2} d\omega'$$

$$n(\omega) - 1 = \frac{2}{\pi} P \int_0^{+\infty} \frac{\omega' \kappa(\omega')}{\omega'^2 - \omega^2} d\omega'$$

$$\kappa(\omega) = -\frac{2\omega}{\pi} P \int_{-\infty}^{+\infty} \frac{n(\omega') - 1}{\omega'^2 - \omega^2} d\omega'$$

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If $\varepsilon_1(\omega)$ or $n(\omega)$ deviate in some frequency range from 1, then there must necessarily be absorption structures somewhere.

if either the real or imaginary part of $\varepsilon(\omega)$, $\tilde{n}(\omega)$ is known over the whole spectral range, then the other part can be calculated.